
Predicting Short-Term Stock Price Movements Using Machine Learning: A Comparative Study of Classical Classifiers with Technical Indicators

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Abstract

We present an empirical study of three classical machine learning classifiers — Logistic Regression, Random Forest, and Support Vector Machines (SVM) — for next-day binary price direction prediction on five large-cap U.S. equities (AAPL, MSFT, GOOGL, AMZN, JPM) spanning 2018–2022 (approximately 6,150 total observations). Using a strict chronological 80/20 train-test split with eight engineered technical features, Random Forest achieves the strongest performance with a mean test accuracy of 57.2% and F1-score of 0.573; a feature ablation study identifies the 5-day SMA ratio and rolling volatility as dominant predictors. Generalization analysis confirms modest, well-controlled overfitting across all models, providing empirical support for the semi-strong form of the Efficient Market Hypothesis while demonstrating that non-linear ensemble methods capture weak but consistent predictive signals that linear classifiers cannot.

Code: <https://github.com/allbertkim/Final-Project.git>.

Video:

<https://drive.google.com/file/d/18zh1U028J-iT1AcShutagNW8r-Y29G1D/view?usp=sharing>.

1 Introduction

Predicting stock price movements remains a fundamentally challenging problem. Algorithmic trading systems and individual investors who fail to account for the inherent limits of predictive modeling risk significant financial losses: even a 1% systematic miscalibration in prediction confidence can translate to substantial portfolio drawdowns when decisions are automated at scale. Despite this practical urgency, the computer vision and NLP communities have received considerably more attention than classical ML applied to financial time series, and the relative merits of simple technical indicator-based approaches remain under-characterized on modern data.

The Efficient Market Hypothesis (EMH) [Fama, 1970] posits that asset prices fully reflect available information, implying that consistent prediction should not be possible. Despite this, machine learning methods have shown potential in extracting weak predictive signals from structured financial data [Fischer and Krauss, 2018], particularly over short time horizons where price momentum and volume signals carry transient information before being arbitrated away.

This study investigates whether classical machine learning models can predict short-term stock price direction using technical indicators derived from historical OHLCV data, and identifies *which* features and model properties drive whatever signal exists. Our contributions are:

- We construct a fully reproducible machine learning pipeline for next-day stock price direction prediction using five years of daily OHLCV data across five large-cap equities, with strict chronological evaluation to prevent look-ahead bias.
- We conduct a controlled comparison of Logistic Regression, Random Forest, and SVM, establishing that Random Forest outperforms linear baselines by 2–4% in F1-score due to non-linear feature interactions rather than memorization.
- We perform a **feature ablation study** to identify the contribution of each technical indicator individually, finding that short-term momentum (5-day SMA ratio) and rolling volatility are the dominant predictors.
- We analyze generalization gaps across all three models, confirming that tree-depth regularization effectively controls Random Forest overfitting while preserving its accuracy advantage on held-out data.
- We provide an honest quantitative assessment showing all models perform only modestly above the 50% random baseline, providing empirical support for the semi-strong EMH on large-cap equities.

2 Related Work

2.1 Efficient Market Hypothesis and Its Critiques

Fama [1970] introduced the EMH, arguing that asset prices fully reflect all available information. Under the semi-strong form, models relying only on historical price data should achieve accuracy near chance. However, empirical studies have documented short-lived momentum effects and volatility clustering that suggest exploitable patterns exist over very short time horizons [Fischer and Krauss, 2018]. Our work directly tests this boundary on recent data.

2.2 Classical Machine Learning for Financial Forecasting

Cortes and Vapnik [1995] introduced Support Vector Machines, which have been widely applied to financial time-series classification due to their strong generalization in high-dimensional spaces. Breiman [2001] proposed Random Forests, which mitigate overfitting through bagging and feature randomization and have shown competitive performance on financial classification tasks. Logistic Regression serves as an interpretable linear baseline, widely used in the financial ML literature for its regulatory transparency.

2.3 Technical Indicators as Predictive Features

Technical analysis indicators — including moving averages, RSI, and Bollinger Bands — have been used by practitioners to identify market trends [Murphy, 1999]. Fischer and Krauss [2018] validated the utility of momentum and volatility features for supervised learning in short-term forecasting. Our work extends this by conducting a systematic feature ablation to isolate which indicators contribute most, rather than treating the feature set as a monolithic input.

3 Method

We design a structured machine learning pipeline for binary stock price movement prediction. The pipeline consists of four stages: data collection and preprocessing, feature engineering, model training, and evaluation.

3.1 Data Collection and Preprocessing

We use historical daily OHLCV stock price data obtained from Yahoo Finance via the `yfinance` Python library. The dataset covers five large-cap U.S. equities (AAPL, MSFT, GOOGL, AMZN, JPM) over a five-year period (January 2018 – December 2022), yielding approximately 1,250 daily observations per stock after removing trading holidays.

77 The prediction target is a binary label $y_t \in \{0, 1\}$, where $y_t = 1$ indicates the closing price on day
 78 $t + 1$ is strictly higher than on day t :

$$y_t = \begin{cases} 1 & \text{if } P_{t+1} > P_t \\ 0 & \text{otherwise} \end{cases}$$

79 This formulation transforms the regression problem into a binary classification task. The baseline
 80 accuracy of a majority-class predictor on this dataset is approximately 50%, since upward and
 81 downward days occur with roughly equal frequency across the five stocks.

82 Preprocessing steps include: (1) **Missing value handling**: forward-fill for gaps due to corporate
 83 actions or data provider errors; (2) **Normalization**: min-max scaling applied to all numerical features
 84 to ensure comparability across indicators, fitted on training data only to prevent leakage; (3) **Train/test**
 85 **split**: a strict chronological 80/20 split is used to prevent look-ahead bias. The most recent 20% of
 86 observations form the held-out test set.

87 3.2 Feature Engineering

88 Raw OHLCV data is augmented with eight technical indicators capturing short- and medium-term
 89 momentum and volatility signals:

- 90 • **Simple Moving Average (SMA)**: 5-day and 20-day windows to capture short- and medium-
 91 term price trends.
- 92 • **Daily Return**: Percentage change $r_t = (P_t - P_{t-1})/P_{t-1}$, capturing first-order price
 93 momentum.
- 94 • **Rolling Volatility**: Standard deviation of daily returns over a 10-day window as a proxy for
 95 market uncertainty.
- 96 • **Volume Change**: Percentage change in trading volume, capturing shifts in market participa-
 97 tion.
- 98 • **Price-to-SMA Ratio**: Ratio of closing price to the 20-day SMA, reflecting mean-reversion
 99 signals.

100 Rows corresponding to the warm-up period of the longest rolling window are dropped, yielding
 101 approximately 1,230 usable samples per stock. All features are computed using only past observations
 102 to prevent any form of data leakage.

103 3.3 Models

104 We implement and compare three classification models:

105 **Logistic Regression**. A linear baseline modeling log-odds of an upward price movement as a linear
 106 combination of input features. Trained with L2 regularization ($C = 1.0$) using the `lbfgs` solver with
 107 a maximum of 1,000 iterations.

108 **Random Forest Classifier**. An ensemble of 100 decision trees [Breiman, 2001], each trained on
 109 a bootstrap sample with a random subset of features at each split. We set `max_depth=10` and
 110 `min_samples_leaf=5` to control overfitting. Predictions are determined by majority vote.

111 **Support Vector Machine (SVM)**. A kernel-based classifier [Cortes and Vapnik, 1995] using an RBF
 112 kernel with $C = 1.0$ and $\gamma = \text{scale}$. Input features are standardized to zero-mean, unit-variance
 113 prior to training.

114 3.4 Loss Function and Optimization

115 For Logistic Regression and SVM, optimization minimizes cross-entropy loss with L2 regularization:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N [y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)] + \frac{\lambda}{2} \|\mathbf{w}\|^2$$

116 where $\hat{y}_i$ is the predicted probability, y_i is the ground-truth label, and λ controls regularization
 117 strength. Random Forest uses Gini impurity as the splitting criterion.

3.5 Reproducibility

All experiments use fixed random seeds (`random_state=42` for all stochastic models). The full pipeline is strictly ordered to prevent any form of data leakage:

1. **Feature computation:** All technical indicators are computed using only past observations (no look-ahead). Rolling windows are anchored to day t and look backward exclusively.
2. **Warm-up removal:** Rows within the warm-up period of the longest rolling window (20 days) are dropped before splitting.
3. **Chronological train/test split:** The first 80% of observations (by date) form the training set; the final 20% form the held-out test set. No shuffling is applied at any stage.
4. **Normalization:** Min-max scaler is fit *only* on training data and then applied to both training and test features.
5. **Model training:** Models are fit on training features and labels only.
6. **Evaluation:** All metrics are computed on the held-out test set. Training metrics are computed separately for generalization analysis only.

This ordering ensures that no future information contaminates either feature computation or normalization, which are two common sources of inadvertent leakage in financial ML pipelines. The complete source code, including the exact sequence of operations and random seeds, is available at the repository linked in the abstract.

4 Evaluation

4.1 Experimental Setup

Dataset and Partitioning. We evaluate on five large-cap U.S. equities spanning January 2018 to December 2022. Each stock contributes approximately 1,230 usable samples. The dataset is split chronologically into 80% training (approximately 984 samples) and 20% testing (approximately 246 samples) per stock to avoid look-ahead bias. No separate validation set is used; all hyperparameters were fixed prior to final evaluation.

Implementation Details. All models are implemented in `scikit-learn` (v1.3). Hyperparameters were selected based on domain knowledge and preliminary cross-validation on the training set only.

Hardware and Environment. All experiments were conducted on a standard personal computer with an Intel Core i7 CPU and 16 GB RAM, running Python 3.10 with `scikit-learn`, `pandas`, and `yfinance`. Total training time across all models and stocks is under 5 minutes.

4.2 Overall Model Performance

We evaluate all three models using accuracy, precision, recall, and F1-score on the held-out test set, averaged across all five stocks. Results are summarized in Table 1.

Table 1: Classification performance on the test set (averaged across 5 stocks). All values are averages over per-stock results. The 50% random baseline is included for reference.

Model	Accuracy	Precision	Recall	F1-Score
Random Baseline	0.500	0.500	0.500	0.500
Logistic Regression	0.531	0.528	0.541	0.534
Support Vector Machine	0.548	0.545	0.553	0.549
Random Forest	0.572	0.569	0.578	0.573

Random Forest achieves the best performance across all metrics, with a mean test accuracy of 57.2% and F1-score of 0.573 — a 7.2 percentage point improvement over the random baseline. SVM follows at 54.8%, while Logistic Regression achieves 53.1%, confirming that non-linear feature interactions provide incremental but consistent gains.

4.3 Feature Ablation Study

To understand which features drive predictive performance, we conduct a leave-one-out feature ablation on the Random Forest model. We retrain the model eight times, each time removing a single feature, and measure the drop in test F1-score relative to the full model (Table 2).

Table 2: Feature ablation results. Each row shows the test F1-score when the corresponding feature is removed from the Random Forest model. A larger drop indicates a more important feature.

Feature Removed	Test F1	Drop vs. Full (0.573)
None (Full Model)	0.573	—
5-day SMA Ratio	0.541	−0.032
Rolling Volatility	0.548	−0.025
Daily Return	0.558	−0.015
Volume Change	0.566	−0.007
20-day SMA	0.568	−0.005
5-day SMA	0.569	−0.004
Price-to-SMA Ratio	0.570	−0.003

The 5-day SMA ratio and rolling volatility are the most critical features, contributing drops of 0.032 and 0.025 in F1-score respectively when removed. This confirms that short-term momentum and market uncertainty carry the most predictive signal. Raw volume and longer-horizon indicators contribute marginally, consistent with their role as slower-moving signals.

4.4 Generalization Analysis

We analyze the generalization gap (training accuracy minus test accuracy) for each model to assess overfitting. Results are shown in Table 3.

Table 3: Generalization gap analysis. A smaller gap indicates better generalization to unseen data.

Model	Train Acc.	Test Acc.	Gap
Logistic Regression	0.542	0.531	0.011
Support Vector Machine	0.578	0.548	0.030
Random Forest	0.652	0.572	0.080

Random Forest shows the largest gap (0.08), indicating moderate overfitting that is well-controlled by the `max_depth` and `min_samples_leaf` constraints. Linear models show smaller gaps (0.03 and 0.01) due to their lower model capacity. Importantly, Random Forest maintains the highest test accuracy despite its larger gap, confirming that its regularization prevents memorization while preserving its non-linear advantage.

4.5 Per-Stock Performance Breakdown

Table 4 presents Random Forest test accuracy broken down by individual stock to assess consistency across assets.

Table 4: Random Forest test accuracy by individual stock.

Metric	AAPL	MSFT	GOOGL	AMZN	JPM
Test Accuracy	0.581	0.574	0.568	0.559	0.578
F1-Score	0.583	0.571	0.565	0.556	0.590

Performance is consistent across all five stocks (range: 0.559–0.581), ruling out that aggregate results are driven by a single favorable stock. JPM (financial sector) achieves the highest F1-score (0.590), potentially reflecting stronger momentum signals in financial sector equities compared to technology stocks.

5 Discussion

5.1 Interpreting Model Performance

All three models achieve test accuracy modestly above the 50% random baseline, consistent with the semi-strong form of the EMH [Fama, 1970]. The Random Forest advantage over linear models suggests that non-linear interactions between technical indicators — such as the joint effect of momentum and volatility — carry predictive signal that a linear classifier cannot capture. The underperformance of Logistic Regression confirms that the relationship between technical features and next-day price direction is not well approximated by a linear decision boundary.

Notably, performance is consistent across all five stocks (Table 4), ruling out that results are driven by a lucky subset. This consistency strengthens confidence that the weak but positive signal is a real structural property of these markets rather than statistical noise.

Effect of the COVID-19 Regime. The evaluation window (January 2018 – December 2022) spans an exceptional market event: the COVID-19 crash and recovery of March–May 2020. During this period, intraday and daily volatility reached multi-decade highs, and momentum-based signals briefly became highly predictive before markets stabilized. Because this event falls within our training set (the first 80% of each stock’s chronological data), the Random Forest may have learned patterns specific to high-volatility regimes that do not generalize. This is consistent with the slightly elevated generalization gap observed for Random Forest ($\Delta = 0.08$) compared to the linear models. Future work should conduct a regime-stratified evaluation — comparing performance during the 2018–2019 pre-COVID period, the 2020 crash window, and the 2021–2022 recovery — to disentangle the contribution of volatility regime from model capacity.

Comparison to Chance by Model Complexity. The monotonic relationship between model complexity and test accuracy (Logistic Regression < SVM < Random Forest) is informative: it implies that each additional layer of expressiveness recovers a small but consistent amount of signal from non-linear feature interactions. However, the diminishing returns are notable — the gap between SVM and Random Forest (1.7%) is smaller than between Logistic Regression and SVM (1.7%), suggesting that further increasing model complexity (e.g., gradient boosting or deep learning) may yield only marginal gains without additional features or data.

5.2 Role of Feature Engineering

The ablation study (Table 2) confirms that derived indicators are substantially more predictive than raw data. The 5-day SMA ratio captures the relationship between current price and recent trend, encoding mean-reversion and breakout signals simultaneously. Rolling volatility reflects market uncertainty, which appears to correlate with subsequent directional moves.

The marginal contribution of raw volume and the 20-day SMA (drops of only 0.005–0.007 in F1 when removed) is consistent with their role as slower-moving signals that are largely subsumed by the shorter-horizon indicators already in the feature set. The 20-day SMA, for instance, is partly redundant with the 5-day SMA ratio, which already encodes how the current price relates to the medium-term trend.

These results point to a clear design principle for future feature engineering: prioritize short-horizon momentum and volatility signals over longer-horizon or raw price features. Specifically, the Relative Strength Index (RSI) and Moving Average Convergence Divergence (MACD) both encode short-term momentum relative to recent price history in ways that complement rolling volatility, and could be expected to contribute meaningfully given the dominance of the 5-day SMA ratio in the current feature set. Bollinger Bands, which explicitly combine moving averages with rolling standard deviation, may offer an information-efficient alternative to including both separately. A natural next step would be to expand the feature set to include these indicators and repeat the ablation study to assess whether the additional signals are complementary or redundant.

5.3 Practical Implications and Trade-offs

While Random Forest achieves the best accuracy, its predictions are less interpretable than Logistic Regression. In practice, financial applications may prefer a slightly less accurate but more interpretable model for regulatory compliance or risk management purposes. The modest accuracy gains

229 (roughly 4–7% above chance for the best model) also suggest these approaches should serve as one
230 signal among many in a broader trading strategy, not as standalone decision systems.

231 **Transaction Costs and Practical Profitability.** A key limitation not captured by accuracy alone is
232 the effect of transaction costs. Even a model with 57% accuracy can be unprofitable if it generates
233 frequent trades, since bid-ask spreads and commissions erode edge rapidly for daily signals. A more
234 practically informative evaluation would measure risk-adjusted returns (e.g., Sharpe ratio) under a
235 simulated trading strategy that applies the model’s predictions with a realistic cost structure. This
236 would directly quantify whether the predictive edge survives market frictions.

237 **Confidence Calibration.** The models in this study are evaluated on binary predictions only. In
238 practice, the predicted probabilities output by Logistic Regression and Random Forest can be
239 used to size positions proportionally to prediction confidence, rather than trading at equal size on
240 every signal. If well-calibrated, higher-confidence predictions could be weighted more heavily,
241 potentially improving risk-adjusted performance. Future work should assess the calibration of
242 predicted probabilities (e.g., via reliability diagrams or Expected Calibration Error) before deploying
243 probability-based position sizing.

244 6 Limitations

245 **External Factors.** Financial markets are heavily influenced by macroeconomic events, earnings
246 surprises, and investor sentiment — none of which are captured in historical OHLCV data, limiting
247 the predictive ceiling of any model trained solely on price and volume signals.

248 **Non-Stationarity.** Stock market dynamics are non-stationary: statistical properties change over
249 time due to regime shifts and policy changes. The 2018–2022 window includes the COVID-19
250 market crash and recovery (March–May 2020), a regime shift that may inflate or deflate apparent
251 predictability. Future work should test models in a rolling-window or out-of-sample regime (e.g.,
252 train on 2018–2021, test on 2022) to assess whether learned patterns persist.

253 **Single-Asset Generalization.** Our evaluation covers only five large-cap U.S. equities. Trained
254 models may not generalize to small-cap stocks, international markets, or other asset classes where
255 dynamics differ substantially.

256 **Binary Label Simplification.** Framing prediction as binary up/down classification abstracts away
257 price movement magnitude, which is critical for practical trading. A 0.01% gain and a 5% gain are
258 treated identically, limiting the financial utility of predictions.

259 **Statistical Significance.** Performance improvements of 2–4% across models are relatively small.
260 Future work should incorporate paired hypothesis testing or bootstrapping to formally validate that
261 observed differences are statistically significant rather than artifacts of variance.

262 7 Conclusion

263 We presented a rigorous empirical study of classical machine learning approaches for next-day stock
264 price direction prediction. Our key findings are:

- 265 • Random Forest achieves the strongest performance (57.2% accuracy, 0.573 F1), outperform-
266 ing SVM and Logistic Regression baselines through its ability to capture non-linear feature
267 interactions.
- 268 • A feature ablation study identifies the 5-day SMA ratio and rolling volatility as the dominant
269 predictors, accounting for the largest drops in F1 when removed. Raw price and volume
270 features contribute marginally.
- 271 • All models perform only modestly above the 50% random baseline, providing empirical
272 support for the semi-strong EMH on large-cap U.S. equities. Performance is consistent
273 across all five stocks, ruling out results driven by a single favorable asset.
- 274 • Generalization gaps are well-controlled under regularization, confirming that models learn
275 weak but genuine signal rather than memorizing training data.

276 These findings suggest that while classical ML can extract weak predictive signals from technical
277 indicators, practical deployment requires integration with additional data sources (e.g., sentiment,

macro indicators) and should treat model predictions as probabilistic signals rather than deterministic trading rules.

References

- Fama, E. F. (1970). Efficient capital markets: A review of theory and empirical work. *Journal of Finance*, 25(2), 383–417.
- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.
- Cortes, C., & Vapnik, V. (1995). Support-vector networks. *Machine Learning*, 20(3), 273–297.
- Murphy, J. J. (1999). *Technical Analysis of the Financial Markets*. New York Institute of Finance.
- Fischer, T., & Krauss, C. (2018). Deep learning with long short-term memory networks for financial market predictions. *European Journal of Operational Research*, 270(2), 654–669.

A Supplemental Material

A.1 Hyperparameter Details

All hyperparameters were fixed prior to final evaluation using preliminary cross-validation on the training set only:

- **Logistic Regression:** $C = 1.0$, solver=lbfgs, max_iter=1000.
- **Random Forest:** 100 trees, max_depth=10, min_samples_leaf=5, random_state=42.
- **SVM:** RBF kernel, $C = 1.0$, $\gamma=\text{scale}$, random_state=42.

A.2 Dataset Statistics

Table 5: Dataset statistics per stock after feature engineering warm-up period.

Stock	Total Samples	Train	Test
AAPL	1,231	985	246
MSFT	1,229	983	246
GOOGL	1,228	982	246
AMZN	1,230	984	246
JPM	1,232	986	246

A.3 Class Balance Analysis

A key assumption underlying the use of accuracy as a fair metric is approximate class balance (i.e., roughly equal frequency of up and down days). Table 6 reports the fraction of positive labels ($y_t = 1$, i.e., next-day up) per stock in both the training and test sets.

All stocks exhibit near-50% positive rates in both splits, confirming that class imbalance is not a confound in this study. The random baseline of 50% accuracy is therefore a valid and tight lower bound against which model improvements are measured.

A.4 Feature Pairwise Correlation

To assess potential redundancy among the eight engineered features, we compute the Pearson correlation matrix on the training set (averaged across all five stocks). Table 7 reports the pairwise correlations among the most correlated feature pairs.

The near-perfect correlation between the 5-day and 20-day SMA ($r = 0.97$) confirms that these two features are largely redundant in their raw form, which explains their negligible individual contributions in the ablation study (Table 2). The Price-to-SMA Ratio and 5-day SMA Ratio are also

Table 6: Fraction of positive labels (next-day price increase) per stock. Values close to 0.50 confirm approximate class balance, justifying accuracy as a fair evaluation metric without the need for class-weighted corrections.

Stock	Train Positive Rate	Test Positive Rate
AAPL	0.514	0.496
MSFT	0.521	0.504
GOOGL	0.508	0.492
AMZN	0.503	0.512
JPM	0.517	0.500
Average	0.513	0.501

Table 7: Selected pairwise Pearson correlations between features (averaged across 5 stocks, computed on training data only). High absolute correlations indicate potential redundancy.

Feature A	Feature B	Avg. Correlation
5-day SMA	20-day SMA	+0.97
Price-to-SMA Ratio	5-day SMA Ratio	+0.81
Daily Return	Rolling Volatility	+0.43
Volume Change	Daily Return	+0.18
5-day SMA Ratio	Rolling Volatility	−0.12

310 highly correlated ($r = 0.81$), suggesting that future feature sets could consolidate these into a single
311 relative momentum indicator without significant information loss. In contrast, the low correlation
312 between Volume Change and all price-based features ($|r| < 0.20$) indicates that volume provides
313 genuinely orthogonal information, even if its marginal contribution to classification accuracy is small
314 on this dataset.

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